

Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES
Topic 04
Multivariate Analysis

Jan-May 2026

Principal Component Analysis

- Concerned with explaining the variance-covariance structure of a set of variables through a few linear combinations of these variables.
- Objectives
 - Data Reduction.
 - Interpretation.
- Serves as intermediate steps in a larger study (Multiple Linear Regression, Cluster Analysis).
- Wide application areas: including image analysis, finance, health sector, cryptography, data-privacy etc.

Population Principal Components

- Let $X_1, X_2, \dots, X_p$ be p -random variables with variance-covariance matrix Σ .
- First we want to find a linear combination of $\mathbf{X} = (X_1, X_2, \dots, X_p)$, say $\mathbf{a}'_1 \mathbf{X}$, such that $Var(\mathbf{a}'_1 \mathbf{X})$ is maximum.
- Note that $Var(\mathbf{a}'_1 \mathbf{X}) = \mathbf{a}'_1 \Sigma \mathbf{a}_1$. Now by multiplying $\mathbf{a}_1$ by a constant, we can increase $\mathbf{a}'_1 \Sigma \mathbf{a}_1$ arbitrarily.
- Therefore, a more precise aim is:

$$\max_{\mathbf{a}_1} Var(\mathbf{a}'_1 \mathbf{X}) = \max_{\mathbf{a}_1} \mathbf{a}'_1 \Sigma \mathbf{a}_1,$$

subject to $\mathbf{a}'_1 \mathbf{a}_1 = 1$

- Thus, 1st principal component = linear combination $\mathbf{a}'_1 \mathbf{X}$ which maximizes $Var(\mathbf{a}'_1 \mathbf{X})$ subject to $\mathbf{a}'_1 \mathbf{a}_1 = 1$
- Next, we want to find another linear combination, say $\mathbf{a}'_2 \mathbf{X}$, such that $Var(\mathbf{a}'_2 \mathbf{X})$ is maximum subject to $\mathbf{a}'_2 \mathbf{a}_2 = 1$ and $Cov(\mathbf{a}'_1 \mathbf{X}, \mathbf{a}'_2 \mathbf{X}) = 0$.
- We proceed in the following manner
 - i^{th} principle component = linear combination $\mathbf{a}'_i \mathbf{X}$ that maximizes $Var(\mathbf{a}'_i \mathbf{X})$ subject to $\mathbf{a}'_i \mathbf{a}_i = 1$ and $Cov(\mathbf{a}'_k \mathbf{X}, \mathbf{a}'_i \mathbf{X}) = 0$ for $k < i$.
- Notice that $Cov(\mathbf{a}'_k \mathbf{X}, \mathbf{a}'_i \mathbf{X}) = \mathbf{a}'_i \Sigma \mathbf{a}_k$, and $Var(\mathbf{a}'_i \mathbf{X}) = \mathbf{a}'_i \Sigma \mathbf{a}_i$.

Theorem: Maximization of Quadratic Forms

Let $B_{p \times p}$ be a symmetric non-negative definite matrix with eigen-values $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ and associated normalized eigen-vectors $e_1, e_2, \dots, e_p$. Then,

$$\max_{x \neq 0} \frac{x' B x}{x' x} = \lambda_1 \left(= \frac{e_1' B e_1}{e_1' e_1} \right) \text{ attained at } x = e_1,$$

$$\min_{x \neq 0} \frac{x' B x}{x' x} = \lambda_p \text{ attained at } x = e_p$$

Moreover,

$$\max_{x \neq 0, x \perp e_1, \dots, e_{k-1}} \frac{x' B x}{x x'} = \lambda_k \text{ attained at } x = e_k, k = 1, 2, \dots, p$$

Theorem

Let $\Sigma_{p \times p}$ be the variance-covariance matrix of $\mathbf{X}_{p \times 1} = (X_1, \dots, X_p)$. Let Σ have the eigen-value-eigen-vector pair $(\lambda_1, e_1), (\lambda_2, e_2), \dots, (\lambda_p, e_p)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Then the i^{th} principal component is given by,

$$Y_i = \mathbf{e}_i' \mathbf{X}, \quad i = 1, 2, \dots, p.$$

With these choices $Var(Y_i) = \lambda_i$ and $Cov(Y_i, Y_j) = 0$ for $i \neq j$. Moreover, if some λ_i are equal, the choices of e_i are not unique, and hence Y_i are not unique.

Theorem

$$\sum_{i=1}^p \text{Var}(X_i) = \sigma_{11} + \sigma_{22} + \dots + \sigma_{pp} = \text{tr}(\Sigma) = \lambda_1 + \lambda_2 + \dots + \lambda_p = \sum_{i=1}^p \text{Var}(Y_i).$$

- Therefore, the proportion of total variance explained by i^{th} principal component is $\frac{\lambda_i}{\lambda_1 + \dots + \lambda_p}$, $i = 1, 2, \dots, p$.
- e_{ik} measures the importance of X_k to the i^{th} principal component.

Theorem

If $Y_1 = e_1' \mathbf{X}, Y_2 = e_2' \mathbf{X}, \dots, Y_p = e_p' \mathbf{X}$ are the principal components obtained from the covariance matrix Σ , then the correlation coefficient between the Y_i and X_k is

$$\rho_{Y_i, X_k} = \frac{e_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}.$$

Principal Components Obtained from Standardized Variables

$$Z_i = \frac{X_i - \mu_i}{\sqrt{\sigma_{ii}}}, i = 1, 2, \dots, p$$

$$\Leftrightarrow \mathbf{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ \vdots \\ \vdots \\ Z_p \end{pmatrix} = V^{-1}(\mathbf{X} - \boldsymbol{\mu}), \text{ where } V = \text{diag}(\sqrt{\sigma_{11}}, \sqrt{\sigma_{22}}, \dots, \sqrt{\sigma_{pp}})$$

- $Var(\mathbf{Z}) = \rho_{p \times p} = \begin{pmatrix} \rho_{11}(=1) & . & . & . & \rho_{1p} \\ . & & & & . \\ . & & & & . \\ . & & & & . \\ \rho_{p1} & . & . & . & \rho_{pp}(=1) \end{pmatrix}$

- $\sum_{i=1}^p Var(Y_i) = p$

- Proportion of standardized population variance due to k^{th} principal component = $\frac{\lambda_k}{p}$, $k = 1, 2, \dots, p$, where λ_k 's are the eigen values of $\rho_{p \times p}$.

Summarizing Sample Variation using Principal Components

- Let $x_1, x_2, \dots, x_n$ denote n independent draws from a p -dimensional population with mean vector μ and variance-covariance matrix Σ .
- The sample mean is $\bar{x}$, the sample variance-covariance matrix S .
- 1st sample principal component = Linear combination $a_1' x_j$ that maximizes the sample variance of $a_1' x_j$ subject to $a_1' a_1 = 1$.

- i^{th} sample principal component = Linear combination $a_i'x_j$ that maximizes the sample variance of $a_i'x_j$ subject to $a_i'a_i = 1$ and zero sample correlation between $a_i'x_j$ and $a_k'x_j$ for all $k < i$.
- Note that sample variance of $a_1'x_j$ is $a_1'Sa_1$. The sample covariance between $a_i'x_j$ and $a_k'x_j$ is $a_i'Sa_k$.

Theorem

Let S be $p \times p$ sample variance-covariance matrix with eigenvalue-eigenvector pair $(\hat{\lambda}_1, \hat{e}_1), \dots, (\hat{\lambda}_p, \hat{e}_p)$ with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_p \geq 0$. Then i^{th} sample principal component is given by

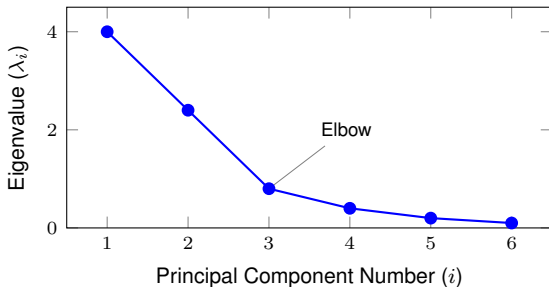
$$\hat{y}_i = \hat{e}_i' \mathbf{x} = \hat{e}_{i1}x_1 + \dots + \hat{e}_{ip}x_p, \quad i = 1, 2, \dots, p$$

Also, sample variance of $\hat{y}_i$ is $\hat{\lambda}_i$, $i = 1, 2, \dots, p$ and sample covariance $\hat{y}_i$ and $\hat{y}_k$ is 0, for $i \neq k$.

Choosing the Number of Principal Components (k)

Objective: Reduce dimensions from p to k ($k \ll p$) while retaining most of the original variation.

- 1 No definite answer.
- 2 **Proportion of Variance:** Retain first k components such that the cumulative variance explained is sufficiently high (typically 70% to 90%).
- 3 **Scree Plot (Elbow Method):** Plot the eigenvalues λ_i (ordered largest to smallest) against their index i . Retain components up to the “elbow” point.



Sample principal components based on standardized data

- Replace S by R , the sample correlation matrix.